

CS 2302 — Data Structures | ABET Course Syllabus | Spring 2026
CRN 27062 | 3 Credit Hours | Prerequisite: CS 1302

Catalog Description:

Abstract data types, algorithm analysis, and their implementation using arrays, linked lists, stacks, queues, trees, and graphs. Emphasis on choosing appropriate data structures and analyzing time/space complexity.

Course Learning Outcomes (mapped to ABET Student Outcomes):

1. Analyze the asymptotic time and space complexity of an algorithm.
2. Implement and evaluate linear data structures (arrays, linked lists, stacks, and queues) for a given problem.
3. Implement and evaluate tree-based and graph-based data structures.
4. Select an appropriate data structure given functional and performance requirements of a problem.
5. Communicate algorithmic trade-offs clearly in written form.

Textbook:

Data Structures and Algorithm Analysis, 3rd Edition, Mark Allen Weiss.

Grading:

Homework 25%, Programming Projects 30%, Midterm 20%, Final Exam 25%.

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Attendance: Attendance is not directly graded but is strongly correlated with performance on programming projects, which build on each other week to week.

Programming Projects: All programming projects must compile and run in the department's standard Linux lab environment. Projects that do not compile receive a maximum of 50% credit, regardless of partial correctness, so test in the lab environment before submitting.

Academic Integrity: Programming projects are individual work unless a project description explicitly states otherwise. Submitting code you did not write, including AI-generated code presented as your own work, is a violation of the university's academic integrity policy.

ABET Assessment Note: Select assignments in this course are used for ABET program assessment purposes. Your work may be reviewed in de-identified form as part of the department's continuous improvement process; this does not affect your grade.

Office Hours: Mondays and Wednesdays, 2:00-3:30 PM, Chemistry & Computer Science Building, Room 214, c